demand for better and cheaper consumer products, such as: cellular telephones, multimedia computers, and high-fidelity music reproduction. These high-revenue applications are shaping the field, while less profitable areas, such as scientific instrumentation, are just riding the wave of technology.

DSPs can be purchased in three forms, as a **core**, as a **processor**, and as a **board level** product. In DSP, the term "core" refers to the section of the processor where the key tasks are carried out, including the data registers, multiplier, ALU, address generator, and program sequencer. A complete **processor** requires combining the core with memory and interfaces to the outside world. While the core and these peripheral sections are designed separately, they will be fabricated on the same piece of silicon, making the *processor* a single integrated circuit.

Suppose you build cellular telephones and want to include a DSP in the design. You will probably want to purchase the DSP as a *processor*, that is, an integrated circuit ("chip") that contains the core, memory and other internal features. For instance, the SHARC ADSP-21060 comes in a "240 lead Metric PQFP" package, only 35×35×4 mm in size. To incorporate this IC in your product, you design a printed circuit board where it will be soldered in next to your other electronics. This is the most common way that DSPs are used.

Now, suppose the company you work for manufactures its own integrated circuits. In this case, you might not want the entire *processor*, just the design of the *core*. After completing the appropriate licensing agreement, you can start making chips that are highly customized to your particular application. This gives you the flexibility of selecting how much memory is included, how the chip receives and transmits data, how it is packaged, and so on. Custom devices of this type are an increasingly important segment of the DSP marketplace.

Lastly, there are several dozen companies that will sell you DSPs already mounted on a printed circuit board. These have such features as extra memory, A/D and D/A converters, EPROM sockets, multiple processors on the same board, and so on. While some of these boards are intended to be used as stand alone computers, most are configured to be plugged into a host, such as a personal computer. Companies that make these types of boards are called **Third Party Developers**. The best way to find them is to ask the manufacturer of the DSP you want to use. Look at the DSP manufacturer's website; if you don't find a list there, send them an e-mail. They will be more than happy to tell you who is using their products and how to contact them.

The present day Digital Signal Processor market (1998) is dominated by four companies. Here is a list, and the general scheme they use for numbering their products: